

Tutorial: Semiconductors in Equilibrium

Fermi Level, Carrier Concentrations, Freeze-Out, and Compensation

Solid State Physics Lessons

October 26, 2025

Reference data (Si, unless stated)

- At $T = 300$ K: band gap $E_g \simeq 1.12$ eV, $n_i \simeq 1.0 \times 10^{10} \text{ cm}^{-3}$.
- Effective density of states: $N_C(300) \approx 2.8 \times 10^{19} \text{ cm}^{-3}$, $N_V(300) \approx 1.04 \times 10^{19} \text{ cm}^{-3}$.
- Temperature scaling (non-degenerate): $N_{C,V}(T) \propto T^{3/2}$, $kT/q \simeq 0.0259$ V at 300 K.
- Intrinsic level: $E_i = \frac{E_C + E_V}{2} + \frac{kT}{2} \ln(N_V/N_C)$.
- Mass-action law (equilibrium): $np = n_i^2$.
- Charge neutrality (full ionisation): $n + N_A = p + N_D$.
- Donor ionisation (with degeneracy $g_D \approx 2$):

$$N_D^+ = \frac{N_D}{1 + g_D e^{(E_F - E_D)/kT}}, \quad n = N_C e^{-(E_C - E_F)/kT}.$$

1 Part A Numerical Problems

Problem 1: Intrinsic Fermi level position vs midgap

Problem Statement. Silicon at 300 K has $N_C = 2.8 \times 10^{19} \text{ cm}^{-3}$, $N_V = 1.04 \times 10^{19} \text{ cm}^{-3}$. Find the offset of the intrinsic Fermi level E_i from the *midgap* position $\frac{E_C + E_V}{2}$.

Problem 2: *n*-type, fully ionised donors at 300 K

Problem Statement. *n*-type Si with $N_D = 1.0 \times 10^{16} \text{ cm}^{-3}$, $n_i = 1.0 \times 10^{10} \text{ cm}^{-3}$. Assuming complete ionisation, find the equilibrium carrier densities n_0 , p_0 and the Fermi level position $E_F - E_i$.

Problem 3: *p*-type, fully ionised acceptors at 300 K

Problem Statement. *p*-type Si with $N_A = 5.0 \times 10^{16} \text{ cm}^{-3}$, $n_i = 1.0 \times 10^{10} \text{ cm}^{-3}$. Find p_0 , n_0 , and $E_i - E_F$.

Problem 4: Lightly n -doped near intrinsic

Problem Statement. Very lightly doped n -Si with $N_D = 1.0 \times 10^{11} \text{ cm}^{-3}$, $n_i = 1.0 \times 10^{10} \text{ cm}^{-3}$. Note that $N_D \approx 10n_i$, so we cannot assume $N_D \gg n_i$. Find n_0 , p_0 , $E_F - E_i$ without approximations.

Problem 5: Compensated semiconductor (full ionisation)

Problem Statement. Si with *both* donors and acceptors: $N_D = 1.0 \times 10^{16} \text{ cm}^{-3}$, $N_A = 3.0 \times 10^{15} \text{ cm}^{-3}$, $n_i = 1.0 \times 10^{10} \text{ cm}^{-3}$. Find n_0 , p_0 , $E_F - E_i$.

Problem 6: Check for degeneracy at high donor density

Problem Statement. Heavily doped Si with $N_D = 5.0 \times 10^{18} \text{ cm}^{-3}$ (fully ionised). Given $N_C = 2.8 \times 10^{19} \text{ cm}^{-3}$, determine whether the non-degenerate (Boltzmann) approximation is valid.

Problem 7: Temperature variation (fully ionised, n_i given)

Problem Statement. n -type Si with $N_D = 1.0 \times 10^{16} \text{ cm}^{-3}$ (fully ionised at all temperatures). At $T = 350 \text{ K}$, use $n_i(350 \text{ K}) = 3.0 \times 10^{11} \text{ cm}^{-3}$ (measured value). Find n_0 , p_0 , and $E_F - E_i$.

Problem 8: Freeze-out in n -Si (incomplete ionisation)

Problem Statement. n -type Si at low temperature ($T = 150 \text{ K}$) with $N_D = 1.0 \times 10^{16} \text{ cm}^{-3}$. Donor ionisation energy: $E_d = E_C - E_D = 45 \text{ meV}$, degeneracy factor $g_D = 2$. At 150 K : $kT = 12.9 \text{ meV}$ and $N_C(150) = 9.9 \times 10^{18} \text{ cm}^{-3}$. Find the ionisation fraction and carrier density (freeze-out regime).

Problem 9: Compensated n -type (full ionisation), Fermi level

Problem Statement. Compensated Si with $N_D = 5.0 \times 10^{16} \text{ cm}^{-3}$, $N_A = 2.0 \times 10^{16} \text{ cm}^{-3}$, $n_i = 1.0 \times 10^{10} \text{ cm}^{-3}$. Find $E_F - E_i$ at 300 K .

Problem 10: Extracting $E_C - E_F$ from Hall-measured n

Problem Statement. Hall measurements give $n_0 = 2.0 \times 10^{15} \text{ cm}^{-3}$ at 300 K . From this data, determine: (i) The distance of Fermi level from conduction band: $E_C - E_F$ (ii) The Fermi level position relative to intrinsic level: $E_F - E_i$

Given: $N_C = 2.8 \times 10^{19} \text{ cm}^{-3}$, $n_i = 1.0 \times 10^{10} \text{ cm}^{-3}$.

2 Part B Conceptual Questions

- Q1. Where is the intrinsic Fermi level in Si at 300 K relative to midgap?
- Q2. What equations determine n_0 and p_0 in equilibrium?
- Q3. How do you compute $E_F - E_i$ in non-degenerate n -type material?
- Q4. What is "freeze-out"?
- Q5. What is a "compensated" semiconductor?
- Q6. When does the non-degenerate (Boltzmann) approximation break down?
- Q7. How do effective masses affect the Fermi level in intrinsic Si?
- Q8. Does $np = n_i^2$ hold in compensated materials?
- Q9. How does temperature affect n_i and the Fermi level in extrinsic n -Si?
- Q10. How to include incomplete ionisation in the neutrality equation?